

LUXOR INTL, Egypt

WMO: 624050

Lat: 25.671N

Lon: 32.707E

Elev: 294

StdP: 14.54

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 42.9	(c) 44.9	(d) 27.2	(e) 21.4	(f) 62.0	(g) 30.0	(h) 24.4	(i) 59.0	(j) 16.5	(k) 64.3	(l) 14.6	(m) 64.7	(n) 1.9	(o) 160	(p) 0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 26.2	(c) 110.8	(d) 74.1	(e) 108.4	(f) 73.3	(g) 106.6	(h) 72.9	(i) 76.7	(j) 104.7	(k) 75.5	(l) 103.7	(m) 74.5	(n) 102.5	(o) 6.5	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.6	102.6	92.9	65.8	96.3	93.2	64.3	91.2	93.3	40.3	104.7	39.1	103.8	38.2	102.8	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.3	(b) 13.3	(c) 11.4	(e) 37.8	(f) 115.4	(g) 2.5	(h) 2.7	(i) 36.0	(j) 117.3	(k) 34.6	(l) 118.9	(m) 33.2	(n) 120.4	(o) 31.4	(p) 122.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.2	59.2	63.2	70.5	79.4	87.3	91.4	92.6	92.3	88.5	81.4	70.4	61.7
	DBStd	13.00	4.40	5.70	6.20	6.39	5.17	3.76	2.94	2.93	3.26	5.04	5.07	4.47
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	435	189	93	17	1	0	0	0	0	0	0	10	125
	CDD50	10304	285	370	634	883	1155	1241	1320	1311	1156	975	612	362
	CDD65	5265	10	43	186	434	690	791	855	846	706	510	172	22
	CDH74	79608	199	670	2345	5859	10653	12965	14224	13826	10479	6297	1786	305
	CDH80	50727	30	224	1042	3393	6952	8898	9866	9480	6667	3481	636	58
(14) Wind	WSAvg	4.3	3.4	4.1	4.7	4.9	4.9	5.2	5.1	4.9	4.3	3.6	3.2	3.0
(15) Precipitation	PrecAvg	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMax	1.3	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	1.2	0.2	0.0
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.9	91.6	100.0	107.9	112.6	113.4	112.1	111.5	108.8	104.3	94.2	85.9
		MCWB	65.1	65.6	69.5	70.7	73.2	74.4	75.4	76.3	74.7	73.4	70.2	66.3
	2%	DB	79.1	86.3	94.7	103.7	108.5	110.1	109.5	108.4	105.9	100.6	89.9	80.7
		MCWB	61.4	63.3	66.5	69.4	71.8	73.3	74.1	74.8	73.5	72.3	68.0	63.1
	5%	DB	75.4	82.1	89.8	99.6	105.5	107.7	107.5	106.2	103.7	97.5	86.5	77.1
		MCWB	59.1	61.1	64.2	67.8	70.4	72.6	73.4	74.0	73.1	71.3	66.3	61.0
	10%	DB	71.8	77.3	85.9	95.6	102.3	105.4	105.4	104.3	100.8	94.8	83.8	73.8
		MCWB	57.1	58.7	62.1	65.7	69.1	71.7	73.0	73.4	72.3	70.5	65.0	59.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.7	67.6	71.7	72.8	74.7	76.5	78.0	78.7	77.8	75.9	72.5	68.9
		MCDB	78.7	87.5	93.4	102.8	107.4	108.3	107.2	105.9	103.7	98.6	89.4	82.3
	2%	WB	62.6	64.6	67.8	70.5	72.9	74.8	76.0	76.7	75.5	73.9	69.5	64.2
		MCDB	77.3	84.7	90.8	99.6	105.3	106.5	105.2	104.7	101.4	96.7	87.0	78.3
	5%	WB	59.7	61.5	65.2	68.5	71.4	73.6	74.8	75.4	74.1	72.4	67.5	61.6
		MCDB	74.0	80.2	87.8	97.1	102.7	104.7	103.7	103.5	99.9	94.8	84.8	75.8
	10%	WB	57.5	59.0	62.7	66.6	69.9	72.4	73.8	74.4	73.0	70.9	65.5	59.6
		MCDB	71.3	76.4	84.7	94.2	100.3	102.6	102.2	102.0	98.5	92.7	82.1	72.9
(35) Mean Daily Temperature Range	MDBR		26.5	27.5	28.2	29.3	28.4	27.7	26.2	26.1	26.7	26.3	26.4	26.2
	5% DB	MCDBR	30.9	32.9	33.1	33.6	32.0	30.5	28.3	27.6	29.0	28.2	28.7	29.2
		MCWBR	18.2	17.7	16.4	15.2	13.9	12.9	11.9	11.5	12.5	13.1	15.0	16.7
	5% WB	MCDBR	30.1	31.9	31.6	32.5	30.0	28.8	26.3	26.4	27.2	26.9	27.8	28.4
		MCWBR	18.3	17.6	16.2	15.1	13.5	12.6	11.6	11.6	12.1	12.9	15.0	16.8
(40) Clear-Sky Solar Irradiance	taub		0.375	0.408	0.451	0.537	0.551	0.473	0.484	0.488	0.469	0.466	0.392	0.372
	taud		2.164	2.045	1.918	1.733	1.721	1.927	1.929	1.926	1.952	1.946	2.138	2.175
	Ebn at Noon		274	273	268	248	244	262	259	257	259	251	266	269
	Edh at Noon		42	50	61	75	75	61	61	61	58	55	43	40
(44) All-Sky Solar Radiation	RadAvg		1440	1735	2088	2345	2481	2591	2509	2367	2147	1807	1516	1349
	RadStd		51	60	63	44	63	27	31	34	47	36	28	24

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.21	+1.08	N/A	+0.67	+0.90	+0.94	N/A	-103	+461	+367
(47) Regional (4 neighbors)		+1.17	N/A	N/A	+0.88	+0.76	+0.74	N/A	-68	+446	+401

Nomenclature: See separate page